Lab 1 – Draft 3

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1 Introduction

Staring at a computer screen all night, whether it is because of gaming, homework, or other reasons, can make it really hard to fall asleep. Not only is it hard to fall asleep, but it is also really easy to lose track of time when staring at a screen for long periods. Due to these factors, coders and gamers are generally known to have awful sleep schedules. This group of individuals struggles with waking up early. Even though their schedule isn't suited for it, they still need to be up for work, class, the gym, etc. This leads to gamers and coders getting a lot less sleep than they need, hence the creation of SleepSync.

1.1 Purpose

SleepSync is an application that helps users fix their sleep schedule. The app accomplishes this by calculating when a user needs to sleep based on their intended wake-up time. Then SleepSync notifies users of their *"wind down"* periods. These function as friendly reminders of what users need to do to start getting ready for bed. Additionally, SleepSync will provide sleep assistance by providing users with helpful external resources for sleep. This could include information on how to fall asleep faster, relaxing music to sleep to, and more.

1.2 Background

SleepSync focuses on the widespread problem of sleep deprivation and how it affects technology users. These include students, shift workers, and other various people who use technology. Sleep deprivation has significant impacts and varying importance on people's productivity, performance, and overall health. While the awareness of sleep debt and overall sleep aid resources has increased over the years, many seem to still struggle to improve their sleep habits due to tools that aren't fit to help. Most of the current market's existing sleep

solutions either require involuntary sleep data entry, a hard setup, or extensive paywalls and monetization. These apps focus more on the actual collection of data and don't provide insights on what should actually be done to help. This creates a gap for an opportunity to create a tool that combines sleep aid education, more personalized recommendations, and other helpful tools that don't intrude on a user's private information if they choose to.

1.3 Scope

The Scope of SleepSync centers on helping users improve their sleep habits through a minimal-input, user-friendly mobile application rather than providing medical diagnosis. The system is intended as a wellness tool that supports better sleep behaviors. To use SleepSync, a user simply enters their intended wake-up time, after which the application calculates recommended bedtimes and provides "wind-down" periods that gently signal when they should begin preparing for sleep. In addition to scheduling guidance, the app offers optional educational and relaxation resources—such as sleep tips, calming audio, and strategies for reducing screen exposure—to further support users in developing healthier habits. SleepSync is designed primarily for students, professionals, shift workers, and other technology users who struggle with consistent sleep schedules, while also serving as a platform that highlights trusted content partners, enriching the user experience and driving visibility to valuable external resources.

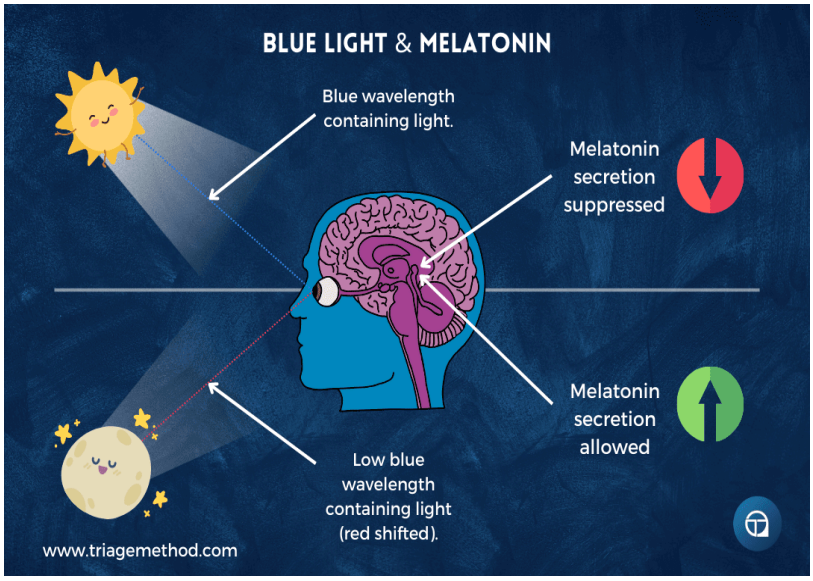

Figure 1: The image pictured describes how blue light affects melatonin.

2 Product Description

Sleep Sync is a personalized app that allows users to track their sleep. It provides an analysis of an individual's sleep quality and offers insight on how to improve it. Some key attributes are: alarm setting, background noises (white, brown, etc.), auto blue light filter, and a sleep score. Users can input personal information to provide a more detailed analysis to their report, such as pregnancy, sex, and age.

Sleep Sync offers a solution to disrupted sleep cycles by encouraging proper sleep habits in a personalized way that doesn't feel meticulous. The objective is to make sleep tracking easy and enjoyable without compromising accuracy. By utilizing detailed information from users, the ideal sleep time can be calculated and offered freely for users.

2.1 Key Product Features and Capabilities

Sleepsync lets users track their sleeping habits and offers resources to users that allow them to work towards healthier habits. Sleepsync uses a formula to calculate things like sleep debt, healthy sleep goals, and more. Sleep is important, and we need a lot of it, but for many people, sleep is not something they reliably get. This has been amplified in recent years with things like short form video content and greater accessibility to technology. SleepSync is unique because our platform provides resources alongside the tools, not just the tools themselves. The key feature that sets us apart is our informative resources, where users can view the reasons for the different health suggestions and learn about exactly why the sleep recommendations are important.

Sleepsync is focused on long term results, meaning that it looks to form healthier habits with sleep that a user will stick to instead of just hollow results that users will abandon within a year. With a set goal of long term growth, SleepSync pushes for small but meaningful changes over time, with access to resources that can explain why these changes are important. By focusing on the health aspect of sleep, the goal is to be as informative as we can be and encourage an understanding of sleep health and why sleep is so important, doing small but meaningful changes over time and ultimately letting the users own motivations guide them with our resources. This type of goal amplifies health and understanding over all, which lead to better results and encourage our users to care about their own health.

2.2 Major Components (Hardware/Software)

The hardware required to support the solution includes a PC or laptop, and potentially, in the future, users will have the option to use their mobile devices and watches. The software to be used includes a PostgreSQL database to store user data, a blue light filter API, a YouTube API, and a Website search API specifically targeting sleep resources. Software we plan on developing includes a sleep calculation tool that calculates the user's sleep schedule and a “wind-down” time. We will also ask for user personal data like their gender, age, and pregnancy to better assess their sleep. User devices will interact with the SleepSync application, which will allow the user to quickly access sleep resources such as videos, articles, and sleep assets like background noise.

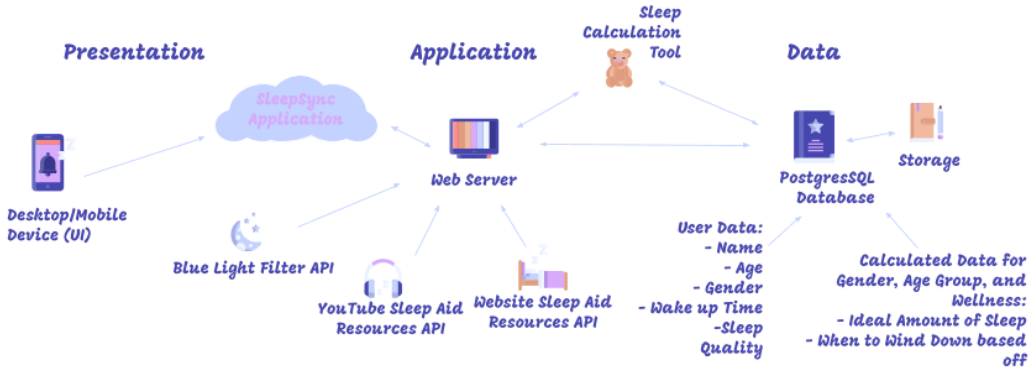

Figure 2: This diagram illustrates how SleepSync’s Presentation, Application, and Data layers work together to deliver a seamless user experience.

Competition Matrix

Key	Feature	SleepSync	Calm	Headspace	SleepCycle	Pillow
Yes	Calculates Bedtime	✓			✗	✓
	Informs Users of "Wind-Downs"	✓	✗	✗	✓	
Partially	Provides Sleep Aid Resources	✓	✗	✗	✗	✗
	Users Can Set a Sleep Schedule	✓	✓	✓	✗	✓
Paywall	Informs User of Sleep Debt	✓			✗	✗
	Relaxing UI	✓	✓	✓	✓	✓
	100% Free Content	✓				

Table 1: Unlike other sleep apps that restrict key features behind paywalls, SleepSync provides every tool, completely free, to help users achieve better sleep without limits.

3 Identification of Case Study

This product is being developed for end-users who want to track their sleep cycles and improve their overall sleep quality, primarily targeting young adults and late teenagers. Improving sleep contributes to better physical and mental health, supports stress management, enhances cognitive function, and assists with healthy weight regulation.

Case study group - a group of 5 students at Old Dominion University. They are a primary user for Sleep Sync, and they are easily available to the developers. Future patrons are parents, rotating/night-shift workers, health-conscious people, counselors recommending Sleep Sync to clients and content creators. 5 ODU students would utilize our prototype application, which will log and calculate their sleep. The purpose of this is to introduce sleep tracking to a small group and receive feedback on any improvements in their sleep quality. Future implementations would include linking articles, YouTube videos, and background noise features in the application.

4 Product Prototype Description

SleepSync is an application that aims to help its users develop more consistent and healthier sleep habits by tracking their sleep, providing resources to improve their sleep quality, and showing insights on a user's sleep habits. The purpose of the prototype is to display how the real world product would function on a smaller, manageable scale. The prototype will have the key features of SleepSync while omitting some of the analytical and content oriented features. SleepSync's prototype will still retain the same look and feel as the real world product while lacking some of the in-depth functionality.

4.1 Prototype Architecture (Hardware/Software)

The SleepSync prototype will be developed using a three-tier architecture, similar to the RWP. The prototype includes a presentation layer, application layer, and data layer. The presentation layer consists of a Desktop/Mobile Device UI that communicates with the SleepSync Application. The application layer is built around a central Web Server that connects to the Sleep Calculation Tool, Audio Tracker Tool, Sleep Tracker Tool, Blue Light Filter API, and YouTube/Website Sleep Aid Resources API. The data layer consists of a PostgreSQL database that stores user data, including name, age, gender, wake-up time, and sleep quality, and calculates wellness metrics such as ideal sleep duration and wind-down timing. The team will continue using GitHub for version control and project management, Discord for group discussions, and Visual Studio Code as the primary development environment.

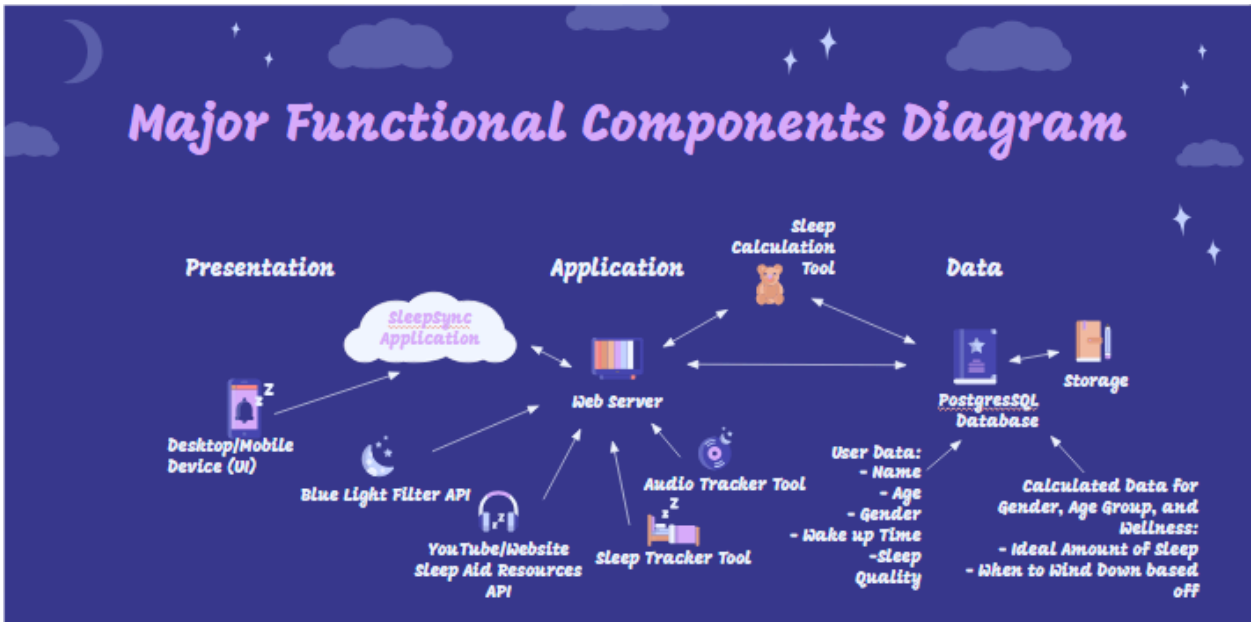

Figure 3: Prototype MFCD

4.2 Prototype Features and Capabilities

Most of the functionality and capabilities of the RWP will be retained for the SleepSync prototype. There will be a limited number of features that will be partially implemented or omitted, as shown in Table 2 below. The features listed illustrate the comparison between the RWP and the prototype and how each is being implemented. The omitted features, Profile Customization, Sleep Debt tracking, Wind-Down Periods, Sleep Resources, Reward Badges, and all Analytics features are excluded from the prototype due to the complexity of their backend dependencies and the time constraints of the development cycle. These features require either extensive data modeling or third-party integrations that fall outside the scope of a functional prototype demonstration.

Category	Feature	RWP	Prototype
Account Management	Account Creation	Fully Implemented	Fully Implemented
	Login/Auth	Fully Implemented	Fully Implemented
	Profile Customization	Fully Implemented	Omitted
	Dashboard	Fully Implemented	Omitted
Sleep Tracking	Bedtime/Wake Up Scheduling	Fully Implemented	Fully Implemented
	Sleep Quality Scoring	Fully Implemented	Fully Implemented
	Sleep Debt	Fully Implemented	Partially Implemented
Bedtime Optimization	Smart Alarm	Fully Implemented	Fully Implemented
	Wind-Down Periods	Fully Implemented	Omitted
	Sleep Resources	Fully Implemented	Omitted
	Reward Badges	Fully Implemented	Omitted
Sleep Report Analytics	By Trends	Fully Implemented	Omitted
	By User Data	Fully Implemented	Omitted
	By Content	Fully Implemented	Omitted

Table 2: RWP vs. Prototype Features

Although some features will not be fully implemented, the final prototype will still allow for a full display as if it were the RWP. A user will be able to create an account, log in securely, input their wake-up time to receive a personalized bedtime recommendation based on sleep cycle calculations, log their nightly sleep to receive a sleep quality score, and set a smart alarm. Together, these core features demonstrate SleepSync's primary value proposition, providing users with an accessible, data-driven tool to improve their sleep health without requiring extensive manual input or configuration.

4.3 Prototype Development Challenges

Numerous risks are identified for the application. They are sorted into three categories: user, customer, and technical. Below are graphs identifying each risk and how it can be mitigated. ** indicates the risk is acknowledged and accepted, as not all risks are able to effectively be managed.

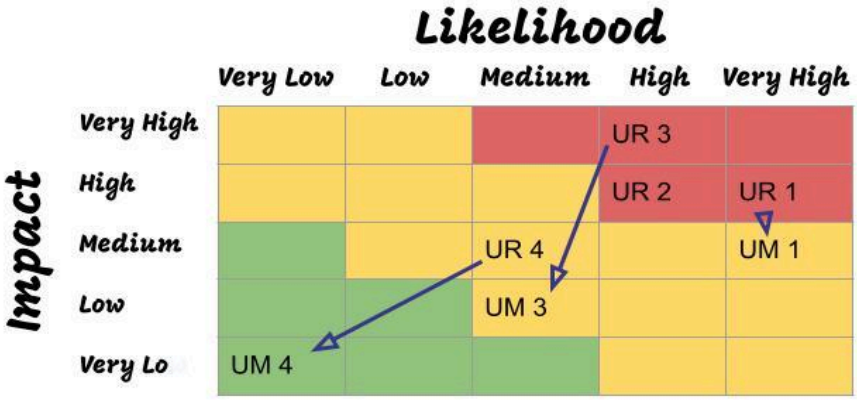

Table 3: User risk matrix for SleepSync application.

Risks:

UR 1 - Data Privacy Concerns (users distrust app w/ personal info)

UR 2 - Users ignoring sleep warnings

UR 3 - User error

UR 4 - Anxiety to perfect sleep

Mitigation:

UM 1 - User legal agreement when signing up

** UR 2 - Risk accepted

UM 3 - Ease of use

UM 4 - Videos and articles regarding stress management

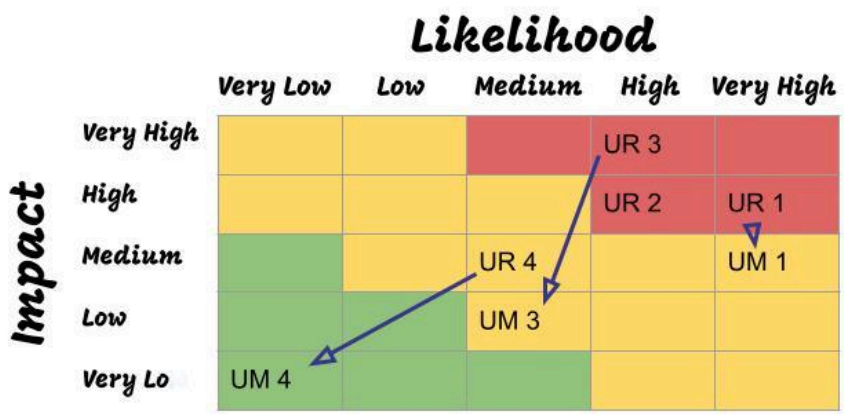

Table 4: Customer risk matrix for SleepSync application.

Risks:

CR 1 - Doubt of effectiveness/improvement

CR 2 - Copyright claims/Legal risks

CR 3 - Lack of professional insight

CR 4 - Lack technical skill

Mitigation:

CM 1 - Trend of improvement over time

CM 2 - Properly credited/cited sources

CM 3 - Citing calculation sources/studies

**CM 4 - Risk accepted

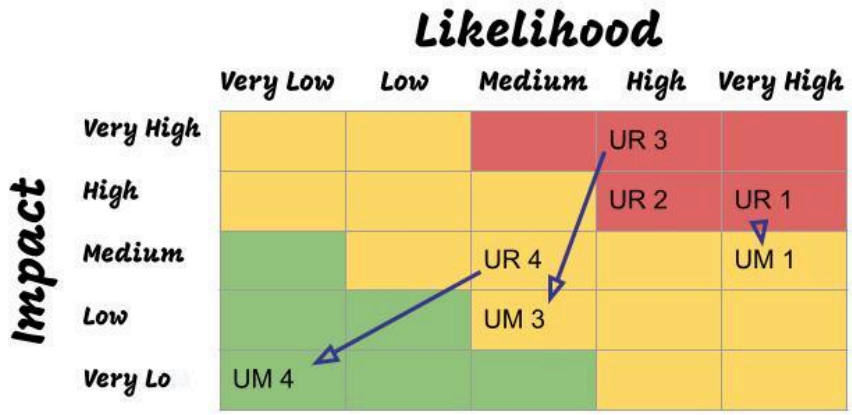

Table 5: Technical risk matrix for SleepSync application.

Risks:

TR 1 - Security Issues(system data could be exposed)

TR 2 - Subjective estimates

TR 3 - No insight to sleep stages

TR 4 - Bugs and Glitches

Mitigation:

TM 1 - Authentication / Encryption, follow best security practices

TM 2 - Sex, age, and activity data

TM 3 - Provide general sleep education

TM 4 - Manual entries/updating frequently

5 Glossary

Sleep Cycle: The repeating pattern of NREM sleep and REM sleep that occurs throughout the night. A full sleep cycle typically lasts 90â€“110 minutes and repeats several times, guiding the body through different stages of brain activity and physical restoration.

REM Sleep: A state of sleep that recurs cyclically with non-REM sleep several times during a normal period of sleep, characterized especially by greatly depressed muscle tone, dreaming with vivid imagery, rapid eye movements, and increased neuronal activity in certain brain regions (such as the pons). It typically comprises up to 25% of time spent in sleep.

NREM Sleep: A state of sleep that occurs regularly during a normal period of sleep with intervening periods of REM sleep and is characterized by delta wave brain activity, minimal dreaming, and a reduced level of autonomic physiological activity.

Blue Light Filter: A screen setting or external tool that reduces the amount of blue-wavelength light emitted by a device, reducing eye strain and minimizing interference with the body's natural melatonin production.

Circadian Rhythm: The internal 24-hour biological clock that regulates sleep, wakefulness, hormone release, body temperature, and other daily physiological processes based on patterns of light and darkness.

Sleep Debt: The cumulative effect of not getting enough sleep over time,leading to fatigue, impaired cognitive performance, and other negative health effects.

Melatonin: A hormone made by the pineal gland (tiny organ near the center of the brain). Melatonin helps control the body’s sleep cycle and is an antioxidant. It is also made in the laboratory and sold as a supplement.

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